Joshua Brown

Address: 4 Winton Road,
Orpington,
Kent
BR6 7AL

Telephone: 07712291964
E-mail: joshua.brown214@imperial.ac.uk

Education

Imperial College London

MEng Electronic and Information Engineering at Imperial College London, 2014 – *present*

Third year specialisms

- Robotics, Concurrent Programming, High Level Programming, Operations Research, Machine Learning, Embedded Systems, Computer Graphics

Second year module highlights

- Computer Architecture (70%), IBM Enterprise Architecture Workshop (67%), Algorithms and Complexity (64%)

First year module highlights

- User-Centered Information Systems (82%), Professional Engineering (72%), Software Engineering 1: Introduction to Computing (72%), Digital Image Processing Project (72%)

Projects

- Built a small scale C to MIPS compiler using C++ with Flex and Bison
- Built a MIPS CPU simulator using C++
- Developed my own single camera 2D-3D image conversion algorithm, and implemented this on an FPGA
- Designed an automated river monitoring system as part of the Nepal Water for Health Challenge *(group project)*
- Designed several innovative browsing interfaces for a news website

Relevant research experience

Undergraduate Research Opportunity - Imperial College London (2016)

- Conducted an investigation into the effects of apparent perspective on Rapid Serial Visual Presentation which will result in a paper for submission in early 2017
- Responsible for implementing experimental interfaces, designing a rigorous experiment, performing that experiment on volunteer subjects and communicating results to other members of my team
- Jointly supervised by Emeritus Professor Robert Spence and Dr. Mark Witkowski, and collaborated with other researchers, including from Mitsubishi Electric Research Labs and Arachnys Information Systems
- Applied for and won the EPSRC vacation bursary to fund my project

Relevant Skills

- Keen design skills, with experience in product and human-computer interface design
- Sound knowledge of programming in C/C++, Python, SQL and several RISC assembly languages, web design in HTML and object oriented software engineering
- Experience working with PIC microcontrollers, the ARM mbed, Raspberry Pi and FPGA platforms
- Excellent CAD skills including Adobe Creative Suite, Solidworks and Autodesk Inventor
- Wide experience of manual and CNC manufacturing, including laser cutting, 3D printing, milling and metal turning

Other education

Wilmington Grammar School for Boys, 2007 - 2014

Qualifications

- A-Levels in Systems and Control (A*), Mathematics (A) and Physics (A) 2014
- AS-Level in Chemistry (A) 2013
- 9 GCSEs at A or A* (A* in English Language, Mathematics and German) 2012
- A* in the Level 2 Diploma in Engineering (full marks in 3 out of 8 units) 2012
- Distinction in the OCR First Certificate in ICT and full marks in the Higher Project Qualification 2012

Projects, 2009-2014

- Built a 3D printer donated by the company – Bits from Bytes – for my school's Design Technology department and trained staff and students to use and maintain it
- Founded and led the school's steam car club
- Helped run the school's stand at the Dartford Industrial Heritage day
- Part of a group of students building a Caterham 7 Kit car in the Caterham '7s for schools' project
- Led a team in the Mission Virgin Galactic Competition, in which we won the South East finals
- Designed and launched a website to showcase the work of my school's Engineering Department
- Designed a metro map style visualization of the School's 20 or so bus routes after concerns that students were getting confused by the multiple operators' routes

Work Experience

Work Shadowing - Rolls-Royce, Derby (2013)

- Witnessed and engaged with the work of a world class engineering company in design, stress analysis, project planning, manufacturing engineering, control systems and maintenance

Work experience – Specac, Orpington (2011)

- Gained practical skills in 2D and 3D design and manual manufacturing

Work experience – UBC/Interbulk, Petts Wood (2011)

- Learned to program in Delphi, implementing a database and GUI for a small contacts organizer
- Learned about the roles of engineers and developers in a business with a non-technical focus

Awards

- Imperial College President's Scholarship 2014
- Bloodhound SSC Award for Excellence in Engineering 2013
- Advanced STEM Leaders Award 2013
- Rolls-Royce Arkwright Engineering Scholarship 2012

Interests and activities

- I currently work as a guide, giving department and campus tours on open days and interview afternoons
- I helped run and organize STEM outreach activities in local primary schools, eventually helping to train other students to continue this work
- I have built several computers, ranging from small form factor workstations to media servers
- I am a member of the Imperial College Robotics Society and Imperial College Rifle and Pistol Club
- I was part of a team representing Bromley in archery at the London Youth Games for seven years

References available on request